

Daniel Eric Pepper

CONTACT INFORMATION	<i>Email: daniel.pepper2@gmail.com</i>	<i>WWW: https://daniel-e-pepper.github.io/</i>
EDUCATION	Ph.D. in Pure Mathematics , <i>York University</i> Advisor: Paul Skoufranis Thesis: Bi-Free Embeddability of Certain Multi-Algebra Independences GPA : 9.0/9.0	<i>2019 - 2026</i>
	MMath in Pure Mathematics , <i>University of Waterloo</i> Advisors: Alexandru Nica and Brian Forrest Project: Basics of Free Stochastic Integration	<i>2017 - 2018</i>
	BMath in Pure Mathematics , <i>University of Waterloo</i>	<i>2013 - 2017</i>
TEACHING EXPERIENCE	Course Instructor <i>Department of Mathematics and Statistics, York University</i> MATH 1013: Applied Calculus I (80+ students) MATH 1300: Differential Calculus with Applications (160+ students)	<i>Fall 2024</i> <i>Fall 2023</i>
	Teaching Assistant <i>Department of Mathematics and Statistics, York University</i> MATH 1013: Applied Calculus I MATH 1014: Applied Calculus II MATH 1090: Introduction to Logic for Computer Science MATH 1200: Introduction to Proofs and Logic MATH 1300: Differential Calculus with Applications MATH 1310: Integral Calculus with Applications MATH 1506: Mathematics I for the Biological and Health Science MATH 1507: Mathematics II for the Biological and Health Sciences MATH 2001: Real Analysis I MATH 2015: Applied Multivariate and Vector Calculus MATH 2200: Extended Problems, Conjectures and Proofs MATH 3001: Real Analysis II	<i>2019 - 2025</i>
	<i>Faculty of Mathematics, University of Waterloo</i> • Includes leading tutorials, grading, exam proctoring and holding office hours. • Provided one-on-one help for students taking first and second-year math courses at the university's Math Help Center.	<i>2017 - 2018</i>
SERVICE	External Catalyst and Strategic Consultant <i>Club Infinity (York University's Undergraduate Math Club)</i> Helped revitalize an inactive club to impacting 50+ maths students through renovation and opening of the club room, organization and hosting of competitions and events, and founding a monthly maths newsletter	<i>Sept. 2025 - Present</i>
	Math Mentor <i>York University</i>	<i>2023 - Present</i>

	Organizer and Host: Math and Stats Grad Student Seminar series <i>Department of Mathematics and Statistics, York University</i>	<i>Sept. 2025 - April 2026</i>
	Served as the sole coordinator for 14 graduate student seminars, including recruiting speakers, managing event advertising, facilitating virtual and in-person sessions, and administering attendance verification for Master's students' seminar credit requirements.	
	Deputy Editor and Contributor <i>The Loop (York University's Math Newsletter)</i>	<i>Sept. 2025 - April 2026</i>
	Panelist: Graduate Studies Panel <i>Department of Mathematics and Statistics, York University</i>	<i>Nov. 2025</i>
	Guest Speaker and Assistant: Teaching Assistant Training <i>Department of Mathematics and Statistics, York University</i>	<i>Sept. 2024</i>
TALKS	A First Encounter with Non-Commutative Probability <i>Math and Stats Grad Student Seminar, York University</i>	<i>Sept. 2025</i>
	How Certain Multi-Algebra Independences can Arise in Bi-Free Probability <i>Probabilistic Operator Algebra Seminar</i> Virtual talk invited by Professor Dan-Virgil Voiculescu at the University of California, Berkeley.	<i>July 2025</i>
	Session on Advances in Operator Algebras and Free Probability <i>Conference on Stochastic Processes and their Applications, Wrocław, Poland</i>	<i>July 2025</i>
	A Combinatorial Proof of the Central Limit Theorem <i>Left to the Reader Seminar, York University</i>	<i>Nov. 2019</i>
	Learning Seminar on Free Stochastic Integration <i>Department of Pure Mathematics, University of Waterloo</i>	<i>Summer 2018</i>
PAPERS	Preprints D. Pepper, <i>How Free-Free-Boolean Independence Arises in Bi-Free Probability</i> , arXiv:2505.18274	
AWARDS AND FELLOWSHIPS	York University Graduate Fellowship	<i>2019 - 2025</i>
	Research Assistantship <i>Department of Mathematics and Statistics, York University</i>	<i>2019 - 2025</i>
	Departmental 3-Minute Thesis Winner <i>Department of Mathematics and Statistics, York University</i>	<i>Feb. 2023</i>
UNDERGRADUATE RESEARCH MENTORSHIP	Distance From Rank- r Projections to Nilpotents <i>Summer Project, NSERC USRA and ENURA, York University</i> Mentees: Joe Tran and Mohammad Azeem. Supervised by Paul Skoufranis.	<i>Summer 2023</i>
OUTREACH AND MATHS COMMUNICATION	Maths Youtube Channel — A Curious Mathematician Dedicated to sharing the beauty of abstract mathematics with a general audience. 2800+ subscribers. Handcrafted Variant Sudoku Puzzles 20 unique puzzles aiming to inspire mathematical thinking through simultaneous creativity and logical deduction.	